Biochemistry Cume

Feb 8, 2002

1. (30 points) Beginning with the kinetic scheme for an enzyme catalyzed reaction

$$E + S \underset{k_{-1}}{\overset{k_1}{\leftrightarrows}} ES \xrightarrow{k_2} E + P$$

derive the Michaelis-Menten Equation describing the relationship between reaction velocity, v, and substrate concentration. For full credit, you must state clearly any assumptions and defintions you use in your derivation.

2. (20 points) Although the Michaelis-Menten Equation indicates that key kinetic parameters in the equation can be determined from a plot of v against [S], this is seldom done. A more common approach is to use a Lineweaver-Burk plot in which 1/v is plotted against 1/[S]. Using a sketch of a Lineweaver-Burk plot, show how one can use it to determine the kinetic parameters from the Michaelis-Menten Equation.

3. (20 points) The Lineweaver-Burk plot is also useful in determining whether an enzyme inhibitor is acting in a competitive or noncompetitive manner. Using sketches, describe how this is done.

4. (30 points) An alternative to the Lineweaver-Burk apporach is the Eadie-Hofstee plot in which v is plotted against v/[S]. Starting with the Michaelis-Menton Equation, do the algebraic transformation to arrive at an equation which describes the Eadie-Hofstee plot. Using sketches, show how such a plot could be used to determine the kinetic parameters from the Michaelis-Menten Equation and also to distinguish between competitive and noncompetitive inhibitors.